



GPS PIPELINE

Quickstart Guide

Requirements

- Compatible with most operating systems: Linux, Windows ([running Linux with WSL2](#)), macOS
- Java 17+ (recommend using [SDKMAN!](#) to install an appropriate version of [Temurin distribution](#))
- [Docker](#) or [Singularity/Apptainer](#)
- Have at least 16GB of RAM and 100GB of free storage

Setup (Internet connection required)

1. Download or Git Clone the pipeline core files from its GitHub Repository
 - a. Download from: <https://github.com/GlobalPneumoSeq/gps-pipeline/releases>
 - b. To clone, run: `git clone https://github.com/GlobalPneumoSeq/gps-pipeline.git`
2. Initialise the pipeline after changing directory (`cd`) into the pipeline directory:
 - a. Using Docker: `./run_pipeline --init`
 - b. Using Singularity: `./run_pipeline --init -profile singularity`
 - c. Add `--annotation` to get Bakta database if you expect to perform annotations (GFF3)
3. It can take a while to download (16GB container images + 12.2GB databases; extra 3.8GB for Bakta database)

Run (No internet connection required after initialisation)

1. Run the pipeline by specifying the directory containing your FASTQ files via `--reads`
 - a. Using Docker: `./run_pipeline --reads /path/to/reads-dir`
 - b. Using Singularity: `./run_pipeline --reads /path/to/reads-dir -profile singularity`
 - c. Add `--annotation` if you want the pipeline to perform annotations (GFF3) as well
2. Grab a cup of tea and wait

Tip 1:

If you have not [specified output path](#) with `--output`, the default is the `output` directory in the pipeline directory.

Tip 2:

Each input sample will generate ~2GB intermediate files on average. You might need to process your samples in batches if the storage space is limited on your system. The `clean_pipeline` [helper script](#) of the pipeline may be useful after each successful run.

Documentation

- GitHub Repository: <https://github.com/GlobalPneumoSeq/gps-pipeline>

Notice

- The current release of the pipeline only works with Illumina paired-end short reads
- Use a specific version of the pipeline to ensure consistent output for the same study